In [2]:
```
!pip install -q nolds antropy

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
from scipy import stats
import requests, json
import warnings
warnings.filterwarnings('ignore')
import antropy, nolds

plt.style.use('dark_background')

PROGRESS_CRITIQUE_HYPOTHESE = 65.0
MODE = 'fast'

CONFIG = {
    'fast': {
        'n_estimations': 1,
        'window_size': 80,
        'stat_test': 'cohen_d',
        'min_per_group': 2,
        'estimator_min_len': 50,
        'mann_whitney': False,
        'bootstrap': False,
    },
    'robust': {
        'n_estimations': 5,
        'window_size': 150,
        'stat_test': 'cliff_delta',
        'min_per_group': 3,
        'estimator_min_len': 80,
        'mann_whitney': True,
        'bootstrap': True,
    }
}
C = CONFIG[MODE]
print(f"=== BIO_CANCER_LOG v1.0 | MODE = {MODE.upper()} | TCGA-BRCA ===")
print(f"  Estimations : {C['n_estimations']} | Fenêtre : {C['window_size']} pa
      f"Test : {C['stat_test']}")
print(f"  Seuil critique (hypothèse) : {PROGRESS_CRITIQUE_HYPOTHESE}e percenti

# ============================================================================
# 0. STATISTIQUES
# ============================================================================

def cohen_d(g1, g2):
    g1, g2 = np.asarray(g1), np.asarray(g2)
    if len(g1) < 2 or len(g2) < 2: return np.nan
    pooled_std = np.sqrt((np.var(g1, ddof=1) + np.var(g2, ddof=1)) / 2.0)
    return 0.0 if pooled_std < 1e-12 else (np.mean(g1) - np.mean(g2)) / pooled
```

```python
def cliff_delta(g1, g2):
    g1, g2 = np.asarray(g1), np.asarray(g2)
    n1, n2 = len(g1), len(g2)
    if n1 < C['min_per_group'] or n2 < C['min_per_group']: return np.nan, np.r
    try:
        U, _ = stats.mannwhitneyu(g1, g2, alternative='two-sided')
        return np.clip((2 * U - n1 * n2) / (n1 * n2), -1, 1), np.nan
    except: return np.nan, np.nan

def separation_score(g1, g2):
    if C['stat_test'] == 'cohen_d':
        d = cohen_d(g1, g2)
        return abs(d) if not np.isnan(d) else np.nan
    else:
        d, _ = cliff_delta(g1, g2)
        return abs(d) if not np.isnan(d) else np.nan

def mw_score(g1, g2):
    if not C['mann_whitney']: return np.nan
    try:
        _, p = stats.mannwhitneyu(g1, g2, alternative='two-sided')
        return -np.log10(max(p, 1e-300))
    except: return np.nan

# ============================================================================
# 1. ACQUISITION — GDC API TCGA-BRCA (détection automatique des champs)
# ============================================================================

def encode_stage(s):
    """
    Encode le stade AJCC en score [0.25–1.0].
    Gère : 'Stage I/IA/IB', 'Stage II/IIA/IIB', 'Stage III/IIIA/IIIB/IIIC', 'S
    """
    if not isinstance(s, str): return np.nan
    s = s.lower().strip()
    if any(x in s for x in ['not reported','unknown','stage x','indeterminate'
                            'not applicable','stage 0']): return np.nan
    if 'iv' in s:  return 1.00
    if 'iii' in s: return 0.75
    if 'ii' in s:  return 0.50
    if 'i' in s:   return 0.25
    return np.nan

def encode_grade(g):
    if not isinstance(g, str): return np.nan
    g = g.lower().strip()
    if any(x in g for x in ['not reported','unknown','gx','not applicable']):
    if 'g4' in g or 'grade 4' in g or 'anaplastic' in g: return 1.00
    if 'g3' in g or 'grade 3' in g: return 0.75
    if 'g2' in g or 'grade 2' in g: return 0.50
    if 'g1' in g or 'grade 1' in g: return 0.25
    # Breast-specific grades
    if 'poorly' in g:      return 0.75
```

```
    if 'moderately' in g:  return 0.50
    if 'well' in g:        return 0.25
    return np.nan

def encode_t_stage(t):
    """
    Encode le stade T (taille tumorale) depuis ajcc_pathologic_t.
    T1 < 2cm → T4 = invasion directe.
    """
    if not isinstance(t, str): return np.nan
    t = t.lower().strip()
    if any(x in t for x in ['tx','t0','tis','not reported']): return np.nan
    if 't4' in t: return 1.00
    if 't3' in t: return 0.75
    if 't2' in t: return 0.50
    if 't1' in t: return 0.25
    return np.nan

def encode_n_stage(n):
    """
    Encode le stade N (envahissement ganglionnaire) depuis ajcc_pathologic_n.
    N0 = aucun ganglion → N3 = envahissement massif.
    """
    if not isinstance(n, str): return np.nan
    n = n.lower().strip()
    if any(x in n for x in ['nx','not reported']): return np.nan
    if 'n3' in n: return 1.00
    if 'n2' in n: return 0.67
    if 'n1' in n: return 0.33
    if 'n0' in n: return 0.00
    return np.nan


def fetch_tcga_brca(n_cases=1000):
    """
    Télécharge les données cliniques TCGA-BRCA depuis GDC.
    Détecte automatiquement le champ de stade disponible dans la réponse API.
    Champs prioritaires : ajcc_pathologic_stage > tumor_stage > ajcc_pathologi
    """
    url = "https://api.gdc.cancer.gov/cases"

    filters = json.dumps({
        "op": "in",
        "content": {
            "field": "project.project_id",
            "value": ["TCGA-BRCA"]
        }
    })

    # On demande tous les champs potentiels — GDC ignorera ceux qui n'existent
    fields = ",".join([
        "case_id",
        "diagnoses.ajcc_pathologic_stage",
```

```
        "diagnoses.tumor_stage",
        "diagnoses.age_at_diagnosis",
        "diagnoses.days_to_death",
        "diagnoses.days_to_last_follow_up",
        "diagnoses.vital_status",
        "diagnoses.tumor_grade",
        "diagnoses.ajcc_pathologic_t",
        "diagnoses.ajcc_pathologic_n",
        "diagnoses.ajcc_pathologic_m",
    ])

    params = {
        "filters": filters,
        "fields":  fields,
        "format":  "json",
        "size":    n_cases
    }

    print(f"[GDC API] Téléchargement TCGA-BRCA ({n_cases} cas)...")
    resp = requests.get(url, params=params, timeout=60)
    resp.raise_for_status()
    hits = resp.json()['data']['hits']

    # Debug : montrer les clés disponibles dans le premier cas avec diagnoses
    for h in hits[:20]:
        if h.get('diagnoses'):
            d0 = h['diagnoses'][0]
            print(f"  [DEBUG] Champs disponibles dans diagnoses : {sorted(d0.k
            break

    records = []
    for hit in hits:
        diags = hit.get('diagnoses', [{}])
        d = diags[0] if diags else {}

        # Stade : priorité ajcc_pathologic_stage > tumor_stage
        stage_raw = (d.get('ajcc_pathologic_stage') or
                     d.get('tumor_stage') or None)

        records.append({
            'case_id':      hit.get('case_id', ''),
            'stage_raw':    stage_raw,
            'age_days':     d.get('age_at_diagnosis', None),
            'days_death':   d.get('days_to_death', None),
            'days_follow':  d.get('days_to_last_follow_up', None),
            'vital_status': d.get('vital_status', None),
            'grade_raw':    d.get('tumor_grade', None),
            't_raw':        d.get('ajcc_pathologic_t', None),
            'n_raw':        d.get('ajcc_pathologic_n', None),
        })

    df = pd.DataFrame(records)
```

```
    # Encodage
    df['stage_score'] = df['stage_raw'].apply(encode_stage)
    df['grade_score'] = df['grade_raw'].apply(encode_grade)
    df['t_score']     = df['t_raw'].apply(encode_t_stage)
    df['n_score']     = df['n_raw'].apply(encode_n_stage)
    df['age_years']   = pd.to_numeric(df['age_days'], errors='coerce') / 365.2
    df['days_death']  = pd.to_numeric(df['days_death'], errors='coerce')
    df['days_follow'] = pd.to_numeric(df['days_follow'], errors='coerce')
    df['is_dead']     = df['vital_status'].str.lower().str.strip().eq('dead')
    df['survival_days'] = df['days_death'].where(df['is_dead'], df['days_follo
    df['survival_days'] = pd.to_numeric(df['survival_days'], errors='coerce').

    # Si stade AJCC absent, utiliser stade T comme proxy de progression
    mask_no_stage = df['stage_score'].isna()
    df.loc[mask_no_stage, 'stage_score'] = df.loc[mask_no_stage, 't_score']

    n_before = len(df)
    df = df.dropna(subset=['stage_score']).reset_index(drop=True)
    print(f"  {n_before} cas totaux → {len(df)} avec stade connu")

    if len(df) == 0:
        raise ValueError(
            "Aucun stade trouvé. Valeurs brutes uniques de stage_raw : "
            f"{df['stage_raw'].unique()[:10]}\n"
            "Vérifiez les champs disponibles dans [DEBUG] ci-dessus."
        )

    print(f"[OK] {len(df)} patientes | "
          f"Stade I:{(df['stage_score']==0.25).sum()} "
          f"II:{(df['stage_score']==0.50).sum()} "
          f"III:{(df['stage_score']==0.75).sum()} "
          f"IV:{(df['stage_score']==1.00).sum()}")
    print(f"     Âge : {df['age_years'].dropna().mean():.1f} ± "
          f"{df['age_years'].dropna().std():.1f} ans | "
          f"Décédées : {df['is_dead'].sum()} ({100*df['is_dead'].mean():.1f}%)
    return df


def extract_cancer_fluxes(df):
    out = {}

    out['Stage_Signal'] = df['stage_score'].fillna(df['stage_score'].median())

    grade = df['grade_score'].fillna(df['grade_score'].median())
    if grade.isna().all(): grade = pd.Series(np.full(len(df), 0.5))
    out['Grade_Signal'] = grade.values

    age_min, age_max = df['age_years'].min(), df['age_years'].max()
    age_norm = (df['age_years'] - age_min) / (age_max - age_min + 1e-10)
    out['Age_Signal'] = age_norm.fillna(0.5).values

    out['Lethality_Signal'] = df['is_dead'].astype(float).values
```

```
    max_surv = df['survival_days'].dropna().max()
    if max_surv and max_surv > 0:
        surv_loss = 1.0 - (df['survival_days'].fillna(df['survival_days'].medi
    else:
        surv_loss = pd.Series(np.zeros(len(df)))
    out['Survival_Loss'] = surv_loss.clip(0, 1).values

    return out


def compute_progression_level(df):
    stage = df['stage_score'].fillna(df['stage_score'].median())
    grade = df['grade_score'].fillna(df['grade_score'].median())
    if grade.isna().all(): grade = pd.Series(np.full(len(df), 0.5))
    composite = (stage * 0.6 + grade * 0.4)
    return np.array([stats.percentileofscore(composite, v) for v in composite]

# ============================================================================
# 2. SCAN DU SEUIL CRITIQUE
# ============================================================================

def scan_seuil_critique(df, fluxes, progress_series, candidats=None):
    if candidats is None:
        candidats = np.arange(40, 86, 2.5)

    W         = C['window_size']
    flux_keys = list(fluxes.keys())
    n_pat     = len(df)
    sort_idx  = np.argsort(progress_series)
    prog_s    = progress_series[sort_idx]
    flux_s    = {k: v[sort_idx] for k, v in fluxes.items()}

    windows = []
    for i in range(n_pat - W + 1):
        windows.append({
            'progress': float(np.mean(prog_s[i:i+W])),
            'data':     {k: flux_s[k][i:i+W] for k in flux_keys}
        })

    if not windows:
        raise ValueError(f"Pas assez de patientes ({n_pat}) pour la fenêtre de

    scores_par_seuil = []
    for seuil in candidats:
        grp_bas  = [w for w in windows if w['progress'] <  seuil]
        grp_haut = [w for w in windows if w['progress'] >= seuil]
        if len(grp_bas) < C['min_per_group'] or len(grp_haut) < C['min_per_gro
            scores_par_seuil.append(np.nan); continue
        sep = []
        for k in flux_keys:
            d = separation_score(
                np.concatenate([w['data'][k] for w in grp_bas]),
                np.concatenate([w['data'][k] for w in grp_haut])
```

```
            )
            if not np.isnan(d): sep.append(d)
        scores_par_seuil.append(np.mean(sep) if sep else np.nan)

    scores_arr = np.array(scores_par_seuil, dtype=float)
    idx_max    = np.nanargmax(scores_arr)
    seuil_opt  = candidats[idx_max]

    fig, ax = plt.subplots(figsize=(10, 4))
    ax.plot(candidats, scores_arr, 'o-', color='#ff66aa', markersize=6)
    ax.axvline(seuil_opt, color='#ff0066', linestyle='--', linewidth=1.5,
               label=f'Seuil optimal empirique = {seuil_opt:.1f}e %ile')
    ax.axvline(PROGRESS_CRITIQUE_HYPOTHESE, color='#00ffcc', linestyle=':', li
               label=f'Hypothèse (Stade II→III) = {PROGRESS_CRITIQUE_HYPOTHESE
    ax.set_xlabel('Seuil candidat (percentile de progression tumorale)', fonts
    ax.set_ylabel('Séparation moyenne des signatures de régime', fontsize=10)
    ax.set_title('BIO_CANCER_LOG v1.0 — Scan du Seuil Critique Tumoral\n'
                 '(le seuil émerge des données TCGA ; ce n\'est pas Tc analyti
                 fontsize=10)
    ax.legend(fontsize=9)
    ax.grid(True, alpha=0.15)
    plt.tight_layout()
    plt.show()

    print(f"\n[Scan du seuil critique]")
    print(f"  Hypothèse initiale (Stade II→III) : {PROGRESS_CRITIQUE_HYPOTHESE
    print(f"  Seuil optimal TCGA-BRCA           : {seuil_opt:.1f}e percentile
          f"(séparation = {scores_arr[idx_max]:.3f})")
    return seuil_opt, candidats, scores_arr, windows

# ============================================================================
# 3. ESTIMATEURS DE COMPLEXITÉ
# ============================================================================

def apply_estimators(series, flux_name=''):
    s = np.asarray(series).flatten()
    min_len = C['estimator_min_len']
    if len(s) < min_len:
        return {k: np.nan for k in ['Higuchi','DFA','Katz','Hurst','SampleEn']
    if flux_name == 'Lethality_Signal' and np.std(s) > 1e-10:
        s = np.diff(s)
    if np.std(s) > 1e-10:
        s = (s - np.mean(s)) / np.std(s)
    if len(s) < min_len:
        return {k: np.nan for k in ['Higuchi','DFA','Katz','Hurst','SampleEn']
    est = {}
    try: est['Higuchi'] = antropy.higuchi_fd(s)
    except: est['Higuchi'] = np.nan
    try: est['DFA'] = 2.0 - nolds.dfa(s)
    except: est['DFA'] = np.nan
    try: est['Katz'] = antropy.katz_fd(s)
    except: est['Katz'] = np.nan
    try: est['Hurst'] = nolds.hurst_rs(s)
```

```
    except: est['Hurst'] = np.nan
    try: est['SampleEn'] = antropy.sample_entropy(s[:min(500, len(s))], order=
    except: est['SampleEn'] = np.nan
    return est

# ============================================================================
# 4. PIPELINE
# ============================================================================

def build_cancer_features(df, fluxes, progress_series, progress_values):
    W = C['window_size']
    n = C['n_estimations']
    tol = max(3.0, (progress_values[1]-progress_values[0])/2.0) if len(progres

    mode_label = (
        "bootstrap sur sous-cohortes (estimation de stabilité statistique)"
        if C['bootstrap'] else "cohorte unique"
    )
    print(f"\n[Extraction des signatures de régime] {n} {mode_label} | Fenêtre

    sort_idx    = np.argsort(progress_series)
    prog_sorted = progress_series[sort_idx]
    flux_sorted = {k: v[sort_idx] for k, v in fluxes.items()}
    flux_keys   = list(fluxes.keys())
    n_pat       = len(df)

    window_data = []
    for i in range(n_pat - W + 1):
        window_data.append({
            'progress': float(np.mean(prog_sorted[i:i+W])),
            'fluxes':   {k: flux_sorted[k][i:i+W] for k in flux_keys}
        })

    rows = []
    for P in tqdm(progress_values, desc="Scan de progression tumorale"):
        matching = [w for w in window_data if abs(w['progress'] - P) <= tol]
        if not matching:
            matching = [min(window_data, key=lambda w: abs(w['progress'] - P))
        row = {'Progress': P, 'N_cohorts': len(matching)}

        for flux in flux_keys:
            if n == 1:
                combined = np.concatenate([w['fluxes'][flux] for w in matching
                ests = apply_estimators(combined, flux_name=flux)
                for k, v in ests.items():
                    row[f"{flux}_{k}_mean"] = v
            else:
                rng = np.random.default_rng(42)
                feat_vals = {nm: [] for nm in ['Higuchi','DFA','Katz','Hurst',
                for _ in range(n):
                    idx = rng.choice(len(matching), size=max(1,len(matching)//
                    sub = np.concatenate([matching[i]['fluxes'][flux] for i in
                    ests = apply_estimators(sub, flux_name=flux)
```

```
                    for k, v in ests.items():
                        if not np.isnan(v): feat_vals[k].append(v)
                for k, vals in feat_vals.items():
                    if vals:
                        row[f"{flux}_{k}_mean"] = np.mean(vals)
                        row[f"{flux}_{k}_std"]  = np.std(vals, ddof=1) if len(
                    else:
                        row[f"{flux}_{k}_mean"] = np.nan
                        row[f"{flux}_{k}_std"]  = np.nan
        rows.append(row)

    df_feat = pd.DataFrame(rows)
    feature_cols = [c for c in df_feat.columns
                    if c not in ['Progress','N_cohorts'] and not c.endswith('_
    return df_feat, feature_cols, window_data

# ============================================================================
# 5. LES 4 QUESTIONS CANCÉROLOGIQUES
# ============================================================================

def compare_groups_cancer(df_feat, feature_cols, g_low, g_high, label=''):
    grp_bas  = df_feat[df_feat['Progress'].between(g_low[0],  g_low[1])]
    grp_haut = df_feat[df_feat['Progress'].between(g_high[0], g_high[1])]
    if len(grp_bas) < C['min_per_group'] or len(grp_haut) < C['min_per_group']
        print(f"  [!] {label} : groupes trop petits ({len(grp_bas)}/{len(grp_h
    scores = {}
    for col in feature_cols:
        if col not in df_feat.columns: continue
        v1 = grp_bas[col].dropna().values
        v2 = grp_haut[col].dropna().values
        if len(v1) >= C['min_per_group'] and len(v2) >= C['min_per_group']:
            scores[col] = separation_score(v1, v2)
    return pd.Series(scores)

def run_cancer_questions(df_feat, feature_cols, window_data, seuil):
    print(f"\n=== 4 Questions Cancérologiques (seuil empirique = {seuil:.1f}e
    r = {}

    print("Q1 : Phase Localisée vs Métastatique (Stade I–II vs III–IV)...")
    r["Q1_Localise_vs_Metastatique"] = compare_groups_cancer(
        df_feat, feature_cols, [0, 40], [70, 100], "Q1")

    print("Q2 : Tension à la Transition EMT (zone de basculement critique)..."
    marge = 8.0
    r["Q2_Tension_Transition_EMT"] = compare_groups_cancer(
        df_feat, feature_cols,
        [seuil - 2*marge, seuil - marge*0.3],
        [seuil + marge*0.3, seuil + 2*marge],
        "Q2"
    )

    print("Q3 : Agressivité Tumorale (Grade + Létalité)...")
    aggr_cols = [c for c in feature_cols
```

```
                if any(x in c for x in ['Grade_Signal','Lethality_Signal','Su
    r["Q3_Agressivite_Tumorale"] = compare_groups_cancer(
        df_feat, aggr_cols, [0, 30], [70, 100], "Q3")

    print("Q4 : Persistance de Régime (hystérésis intra-cohorte)...")
    flux_names = ['Stage_Signal','Grade_Signal','Age_Signal','Lethality_Signal
    q4 = {}
    for col in feature_cols:
        flux = next((f for f in flux_names if col.startswith(f)), None)
        if flux is None: continue
        effects = []
        for w in window_data:
            s = w['fluxes'].get(flux)
            if s is None or len(s) < 20: continue
            mid = len(s) // 2
            d = separation_score(s[:mid], s[mid:])
            if not np.isnan(d): effects.append(abs(d))
        q4[col] = np.mean(effects) if effects else np.nan
    r["Q4_Persistance_Regime"] = pd.Series(q4)

    return pd.DataFrame({k: v.round(3) for k, v in r.items()}), r

# ============================================================================
# 6. VISUALISATION
# ============================================================================

def plot_clinical_overview(df, progress_series, seuil):
    fig, axes = plt.subplots(3, 2, figsize=(16, 10))

    stage_labels = {0.25:'I', 0.50:'II', 0.75:'III', 1.00:'IV'}
    stage_counts = df['stage_score'].value_counts().sort_index()
    colors_s = ['#00ffcc','#ffcc00','#ff6600','#ff0066']
    axes[0,0].bar([stage_labels.get(s,'?') for s in stage_counts.index],
                  stage_counts.values, color=colors_s[:len(stage_counts)], alp
    axes[0,0].set_title('Distribution des Stades Tumoraux (AJCC)', fontsize=10
    axes[0,0].set_ylabel('Patientes', fontsize=9)
    axes[0,0].grid(True, alpha=0.15)

    grade_labels = {0.25:'G1', 0.50:'G2', 0.75:'G3', 1.00:'G4'}
    grade_counts = df['grade_score'].dropna().value_counts().sort_index()
    if len(grade_counts):
        axes[0,1].bar([grade_labels.get(g,'?') for g in grade_counts.index],
                      grade_counts.values, color='#ff66aa', alpha=0.8)
    axes[0,1].set_title('Distribution des Grades Histologiques', fontsize=10)
    axes[0,1].set_ylabel('Patientes', fontsize=9)
    axes[0,1].grid(True, alpha=0.15)

    for s_val, s_lab, col in zip([0.25,0.50,0.75,1.00],['I','II','III','IV'],
        grp = df[df['stage_score']==s_val]['age_years'].dropna()
        if len(grp): axes[1,0].hist(grp, bins=20, alpha=0.5,
                                    label=f'Stade {s_lab}', color=col)
    axes[1,0].set_title('Âge au Diagnostic par Stade', fontsize=10)
    axes[1,0].set_xlabel('Âge (années)', fontsize=9)
```

```
    axes[1,0].legend(fontsize=8)
    axes[1,0].grid(True, alpha=0.15)

    for s_val, s_lab, col in zip([0.25,0.50,0.75,1.00],['I','II','III','IV'],
        grp = df[df['stage_score']==s_val]['survival_days'].dropna() / 365.25
        if len(grp): axes[1,1].hist(grp, bins=20, alpha=0.5,
                                    label=f'Stade {s_lab}', color=col)
    axes[1,1].set_title('Survie par Stade (années)', fontsize=10)
    axes[1,1].set_xlabel('Survie (années)', fontsize=9)
    axes[1,1].legend(fontsize=8)
    axes[1,1].grid(True, alpha=0.15)

    sc = axes[2,0].scatter(progress_series, df['survival_days']/365.25,
                           c=progress_series, cmap='RdYlGn_r', s=8, alpha=0.6)
    axes[2,0].axvline(seuil, color='#ff0066', linestyle='--', alpha=0.9,
                      label=f'Seuil EMT ({seuil:.1f}e %ile)')
    axes[2,0].axvline(PROGRESS_CRITIQUE_HYPOTHESE, color='#00ffcc',
                      linestyle=':', alpha=0.7, label=f'Hypothèse ({PROGRESS_C
    axes[2,0].set_xlabel('Score de Progression (%ile)', fontsize=9)
    axes[2,0].set_ylabel('Survie (années)', fontsize=9)
    axes[2,0].set_title('Progression vs Survie', fontsize=10)
    axes[2,0].legend(fontsize=8); axes[2,0].grid(True, alpha=0.15)
    plt.colorbar(sc, ax=axes[2,0])

    axes[2,1].hist(progress_series, bins=40, color='#ff66aa', alpha=0.7)
    axes[2,1].axvline(seuil, color='#ff0066', linestyle='--', linewidth=1.5,
                      label=f'Seuil EMT ({seuil:.1f}e %ile)')
    axes[2,1].axvline(PROGRESS_CRITIQUE_HYPOTHESE, color='#00ffcc', linestyle=
                      label=f'Hypothèse ({PROGRESS_CRITIQUE_HYPOTHESE}e %ile)'
    axes[2,1].set_xlabel('Score de Progression (%ile)', fontsize=9)
    axes[2,1].set_ylabel('Nombre de fenêtres', fontsize=9)
    axes[2,1].set_title('Distribution du Paramètre de Contrôle', fontsize=10)
    axes[2,1].legend(fontsize=8); axes[2,1].grid(True, alpha=0.15)

    plt.suptitle('BIO_CANCER_LOG v1.0 — TCGA-BRCA : Vue d\'ensemble Clinique',
    plt.tight_layout()
    plt.show()

def plot_flux_profiles(df, fluxes, progress_series):
    sort_idx = np.argsort(progress_series)
    prog_s   = progress_series[sort_idx]
    flux_meta = [
        ('Stage_Signal',     '#ff66aa', 'Stade Tumoral          [≈ Magnétisati
        ('Grade_Signal',     '#ff6600', 'Grade Histologique     [≈ Énergie E]'
        ('Age_Signal',       '#ffcc00', 'Âge au Diagnostic      [≈ Variance lo
        ('Lethality_Signal', '#ff0066', 'Létalité Observée      [≈ Taux de fli
        ('Survival_Loss',    '#aa44ff', 'Perte de Survie        [≈ Clusters]')
    ]
    fig, axes = plt.subplots(5, 1, figsize=(16, 12), sharex=True)
    for ax, (flux, color, label) in zip(axes, flux_meta):
        vals = fluxes[flux][sort_idx]
        smooth = pd.Series(vals).rolling(30, min_periods=5, center=True).mean(
        ax.scatter(prog_s, vals, color=color, s=2, alpha=0.2)
```

```
        ax.plot(prog_s, smooth, color=color, linewidth=1.5)
        ax.set_ylabel(label, fontsize=8)
        ax.axhline(0, color='gray', alpha=0.3, linewidth=0.5)
        ax.grid(True, alpha=0.1)
    axes[-1].set_xlabel('Score de Progression (%ile)', fontsize=9)
    plt.suptitle('BIO_CANCER_LOG v1.0 — Les 5 Flux Biologiques TCGA-BRCA\n'
                 '(patientes ordonnées par progression Stage×Grade)', fontsize
    plt.tight_layout()
    plt.show()

def plot_landscape(df_feat, seuil):
    mean_cols = [c for c in df_feat.columns if c.endswith('_mean')]
    n = min(len(mean_cols), 15)
    fig, axes = plt.subplots(3, 5, figsize=(20, 10))
    axes = axes.flatten()
    for i, col in enumerate(mean_cols[:n]):
        ax = axes[i]
        ax.plot(df_feat['Progress'], df_feat[col], 'o-', color='#ff66aa', mark
        std_col = col.replace('_mean','_std')
        if std_col in df_feat.columns:
            ax.fill_between(df_feat['Progress'],
                            df_feat[col]-df_feat[std_col],
                            df_feat[col]+df_feat[std_col],
                            alpha=0.2, color='#ff66aa')
        ax.axvline(seuil, color='#ff0066', linestyle='--', alpha=0.8,
                   label=f'Seuil ({seuil:.1f})')
        ax.axvline(PROGRESS_CRITIQUE_HYPOTHESE, color='#00ffcc', linestyle=':'
        ax.set_title(col.replace('_mean','').replace('_',' '), fontsize=7)
        ax.set_xlabel('Progression (%ile)', fontsize=7)
        ax.grid(True, alpha=0.15)
        if i == 0: ax.legend(fontsize=7)
    for j in range(n, len(axes)):
        axes[j].set_visible(False)
    plt.suptitle(
        f"BIO_CANCER_LOG v1.0 [{MODE}] — Paysage de Complexité TCGA-BRCA\n"
        f"(seuil empirique = {seuil:.1f}e %ile | hypothèse EMT = {PROGRESS_CRI
        fontsize=11)
    plt.tight_layout()
    plt.show()

def plot_heatmap(summary, seuil):
    fig, ax = plt.subplots(figsize=(12, max(5, len(summary)*0.35)))
    data = summary.fillna(0).copy()
    for c in data.columns:
        m = data[c].max()
        if m > 0: data[c] /= m
    im = ax.imshow(data.values, aspect='auto', cmap='magma', vmin=0, vmax=1)
    ax.set_xticks(range(len(data.columns)))
    ax.set_xticklabels(data.columns, rotation=25, ha='right', fontsize=9)
    ax.set_yticks(range(len(data.index)))
    ax.set_yticklabels([idx.replace('_',' ') for idx in data.index], fontsize=
    for i in range(len(data.index)):
        for j in range(len(data.columns)):
```

```
            v = summary.iloc[i, j]
            if not np.isnan(v):
                ax.text(j, i, f"{v:.2f}", ha='center', va='center',
                        color='black' if data.iloc[i,j]>0.6 else 'white', font
    plt.colorbar(im, ax=ax, label="Score de séparation normalisé")
    plt.title(
        f"BIO_CANCER_LOG v1.0 [{MODE}] — Heatmap des Signatures de Régime Tumo
        f"TCGA-BRCA | Seuil critique empirique = {seuil:.1f}e percentile", for
    plt.tight_layout()
    plt.show()

# ============================================================================
# 7. EXÉCUTION PRINCIPALE
# ============================================================================

PROGRESS_SCAN = np.round(np.concatenate([
    np.arange(5,  45, 5.0),
    np.arange(45, 78, 2.0),
    np.arange(78, 96, 5.0)
]), 1)

print(f"\nGradient de progression : {len(PROGRESS_SCAN)} points")
print("=" * 70)

print("\n[1/6] Téléchargement TCGA-BRCA depuis GDC API...")
df = fetch_tcga_brca(n_cases=1000)

print("\n[2/6] Extraction des flux biologiques et du score de progression...")
fluxes          = extract_cancer_fluxes(df)
progress_series = compute_progression_level(df)
print(f"  Score de progression : min={progress_series.min():.1f} | "
      f"max={progress_series.max():.1f} | mean={progress_series.mean():.1f}")

print("\n[3/6] Scan du seuil critique (40–85e percentile)...")
seuil_opt, candidats_scan, scores_scan, windows_scan = \
    scan_seuil_critique(df, fluxes, progress_series)
PROGRESS_CRITIQUE = seuil_opt

print("\n[4/6] Vue d'ensemble clinique TCGA-BRCA...")
plot_clinical_overview(df, progress_series, PROGRESS_CRITIQUE)

print("      Profils des 5 flux biologiques...")
plot_flux_profiles(df, fluxes, progress_series)

print("\n[5/6] Extraction des signatures de régime...")
df_feat, feature_cols, window_data = build_cancer_features(
    df, fluxes, progress_series, PROGRESS_SCAN)
print(f"  Signatures extraites : {len(feature_cols)}")
print(df_feat[['Progress','N_cohorts']+feature_cols[:4]].head(3).to_string())

plot_landscape(df_feat, PROGRESS_CRITIQUE)

print("\n[6/6] 4 Questions cancérologiques...")
```

```python
summary, results = run_cancer_questions(
    df_feat, feature_cols, window_data, PROGRESS_CRITIQUE)
print("\n>>> RÉSULTATS — Scores de séparation des signatures de régime :")
print(summary.to_string())

print("\n>>> TOP 3 DES SIGNATURES DE RÉGIME PAR QUESTION :")
for q in summary.columns:
    top = summary[q].dropna().sort_values(ascending=False).head(3)
    if len(top):
        print(f"\n  [{q}]")
        for f, v in top.items():
            print(f"    {f.replace('_mean',''):44s} : Score = {v:.3f}")

plot_heatmap(summary, PROGRESS_CRITIQUE)
print(f"\n[OK] BIO_CANCER_LOG v1.0 terminé.")
print(f"     {len(df)} patientes TCGA-BRCA | "
      f"Seuil EMT empirique : {PROGRESS_CRITIQUE:.1f}e percentile")
```

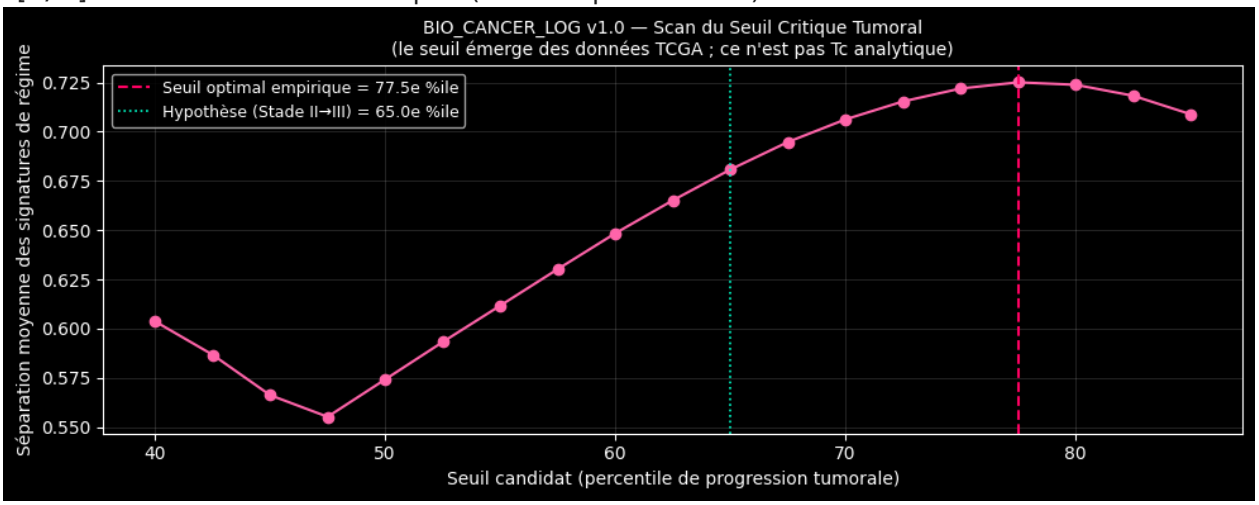
```
=== BIO_CANCER_LOG v1.0 | MODE = FAST | TCGA-BRCA ===
  Estimations : 1 | Fenêtre : 80 patientes | Test : cohen_d
  Seuil critique (hypothèse) : 65.0e percentile

Gradient de progression : 29 points
================================================================================

[1/6] Téléchargement TCGA-BRCA depuis GDC API...
[GDC API] Téléchargement TCGA-BRCA (1000 cas)...
  [DEBUG] Champs disponibles dans diagnoses : ['age_at_diagnosis', 'ajcc_pathol
ogic_m', 'ajcc_pathologic_n', 'ajcc_pathologic_stage', 'ajcc_pathologic_t', 'da
ys_to_last_follow_up', 'tumor_grade']
  1000 cas totaux → 913 avec stade connu
[OK] 913 patientes | Stade I:162 II:530 III:201 IV:20
     Âge : 58.9 ± 13.2 ans | Décédées : 0 (0.0%)

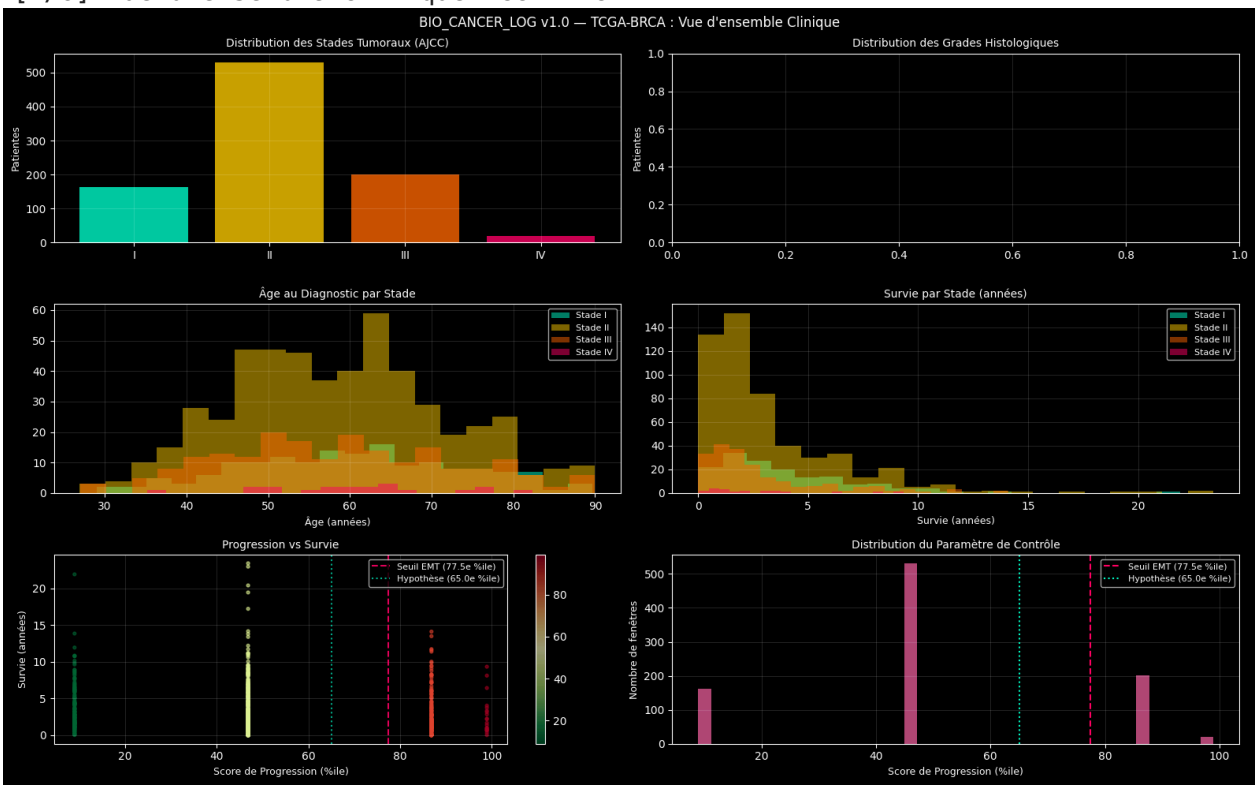
[2/6] Extraction des flux biologiques et du score de progression...
  Score de progression : min=8.9 | max=99.0 | mean=50.1

[3/6] Scan du seuil critique (40–85e percentile)...
```

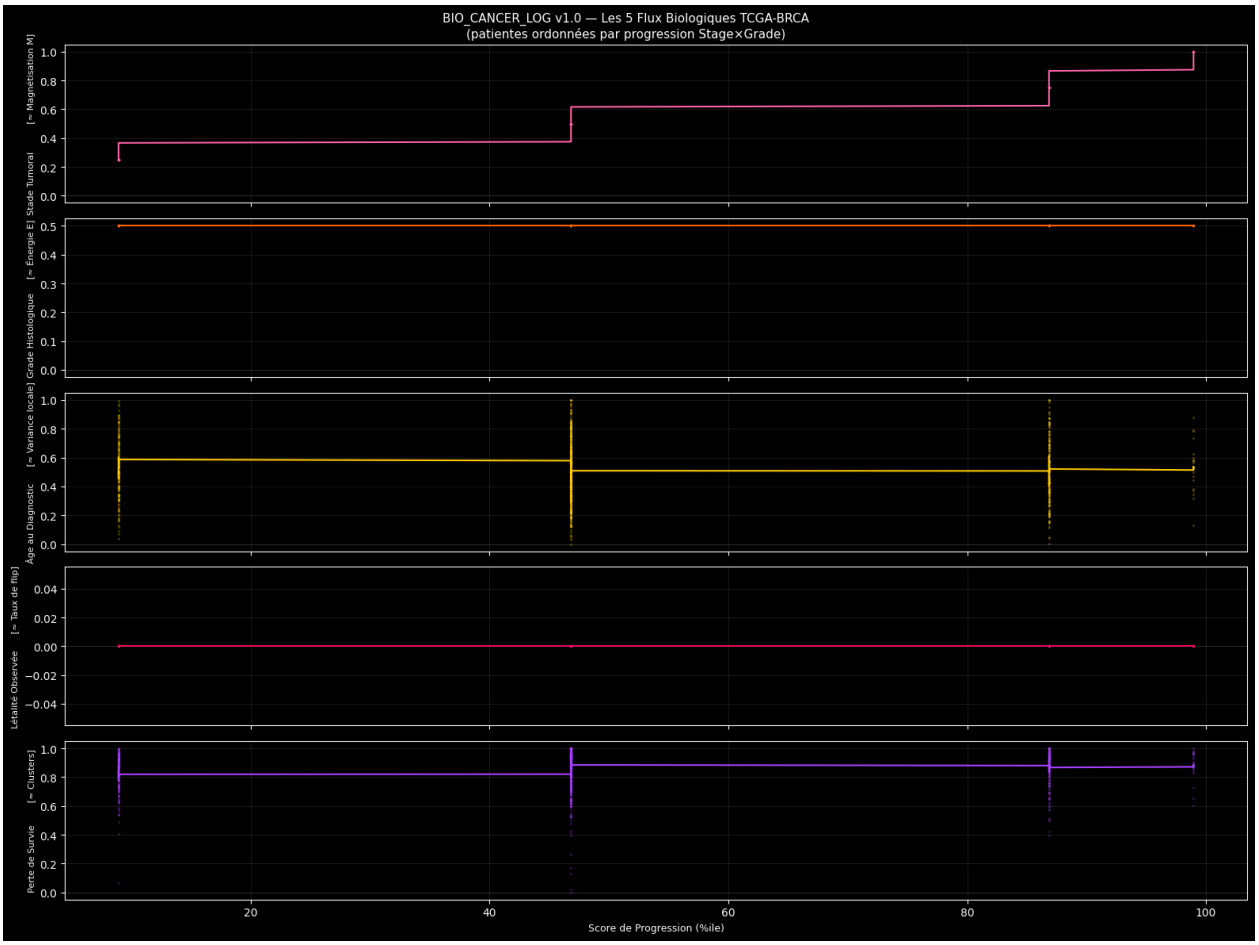


```
[Scan du seuil critique]
  Hypothèse initiale (Stade II→III) : 65.0e percentile
  Seuil optimal TCGA-BRCA           : 77.5e percentile (séparation = 0.725)

[4/6] Vue d'ensemble clinique TCGA-BRCA...
```

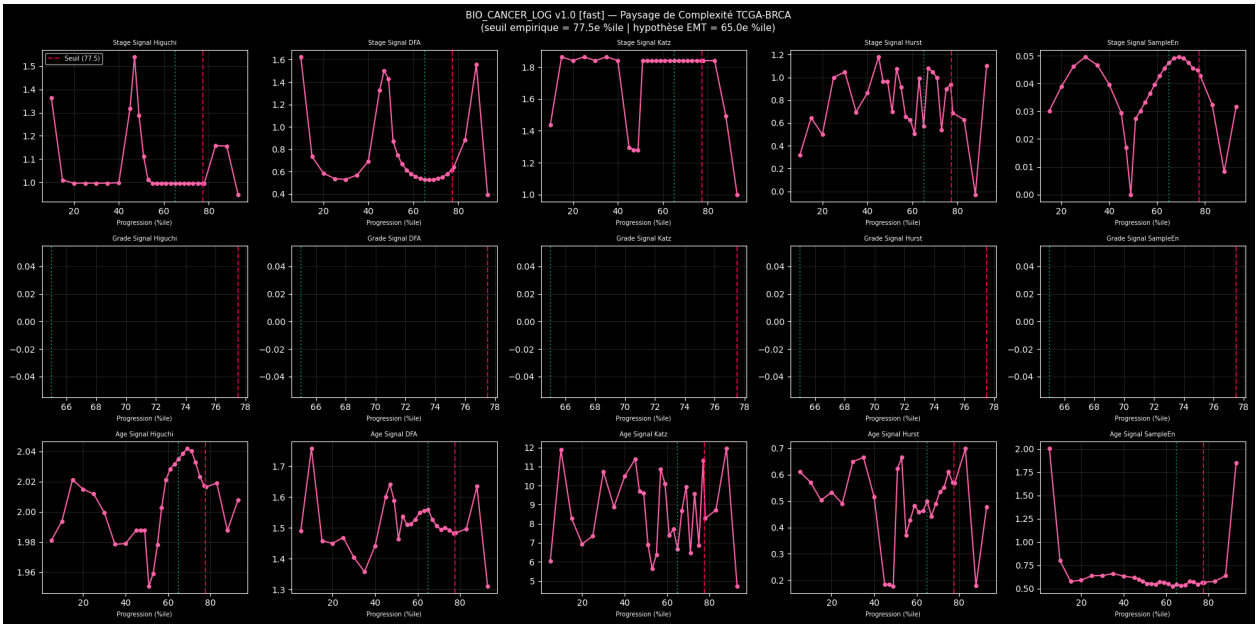



```
  Profils des 5 flux biologiques...
```




```
[5/6] Extraction des signatures de régime...

[Extraction des signatures de régime] 1 cohorte unique | Fenêtre=80 patientes
Scan de progression tumorale: 100%|██████████| 29/29 [02:35<00:00,  5.36s/it]
  Signatures extraites : 25
   Progress  N_cohorts  Stage_Signal_Higuchi_mean  Stage_Signal_DFA_mean  Stag
e_Signal_Katz_mean  Stage_Signal_Hurst_mean
0       5.0          1                        NaN                    NaN
NaN                      NaN
1      10.0         91                   1.365302               1.624911
1.437863                 0.318400
2      15.0         13                   1.009284               0.732857
1.863632                 0.644915
```


BIO_CANCER_LOG v1.0 [fast] — Paysage de Complexité TCGA-BRCA
(seuil empirique = 77.5e %ile | hypothèse EMT = 65.0e %ile)
Stage Signal Higuchi
Seuil (77.5)
Stage Signal DFA
Stage Signal Katz
Stage Signal Hurst
Stage Signal SampleEn
Grade Signal Higuchi
Grade Signal DFA
Grade Signal Katz
Grade Signal Hurst
Grade Signal SampleEn
Age Signal Higuchi
Age Signal DFA
Age Signal Katz
Age Signal Hurst
Age Signal SampleEn
Progression (%ile)

```
[6/6] 4 Questions cancérologiques...

=== 4 Questions Cancérologiques (seuil empirique = 77.5e %ile) ===
Q1 : Phase Localisée vs Métastatique (Stade I–II vs III–IV)...
Q2 : Tension à la Transition EMT (zone de basculement critique)...
Q3 : Agressivité Tumorale (Grade + Létalité)...
Q4 : Persistance de Régime (hystérésis intra-cohorte)...

>>> RÉSULTATS – Scores de séparation des signatures de régime :
                              Q1_Localise_vs_Metastatique  Q2_Tension_Transit
ion_EMT  Q3_Agressivite_Tumorale  Q4_Persistance_Regime
Age_Signal_DFA_mean                                 0.084
0.334                       NaN                 0.191
Age_Signal_Higuchi_mean                             1.246
2.477                       NaN                 0.191
Age_Signal_Hurst_mean                               0.367
0.326                       NaN                 0.191
Age_Signal_Katz_mean                                0.148
0.170                       NaN                 0.191
Age_Signal_SampleEn_mean                            0.173
0.934                       NaN                 0.191
Grade_Signal_DFA_mean                                 NaN
NaN                         NaN                 0.000
Grade_Signal_Higuchi_mean                             NaN
NaN                         NaN                 0.000
Grade_Signal_Hurst_mean                               NaN
NaN                         NaN                 0.000
Grade_Signal_Katz_mean                                NaN
NaN                         NaN                 0.000
Grade_Signal_SampleEn_mean                            NaN
NaN                         NaN                 0.000
Lethality_Signal_DFA_mean                             NaN
NaN                         NaN                 0.000
Lethality_Signal_Higuchi_mean                         NaN
NaN                         NaN                 0.000
Lethality_Signal_Hurst_mean                           NaN
NaN                         NaN                 0.000
Lethality_Signal_Katz_mean                            NaN
NaN                         NaN                 0.000
Lethality_Signal_SampleEn_mean                        NaN
NaN                         NaN                 0.000
Stage_Signal_DFA_mean                               0.088
0.977                       NaN                   0.379
Stage_Signal_Higuchi_mean                           0.184
1.062                       NaN                   0.379
Stage_Signal_Hurst_mean                             0.013
0.714                       NaN                   0.379
Stage_Signal_Katz_mean                              0.416
1.326                       NaN                   0.379
Stage_Signal_SampleEn_mean                          0.375
2.418                       NaN                   0.379
Survival_Loss_DFA_mean                              1.068
0.477                     1.146                   0.170
Survival_Loss_Higuchi_mean                          1.500
```

```
0.721                    2.902                  0.170
Survival_Loss_Hurst_mean                              0.997
0.137                    1.039                  0.170
Survival_Loss_Katz_mean                               0.664
0.706                    0.705                  0.170
Survival_Loss_SampleEn_mean                           0.223
0.772                    0.308                  0.170

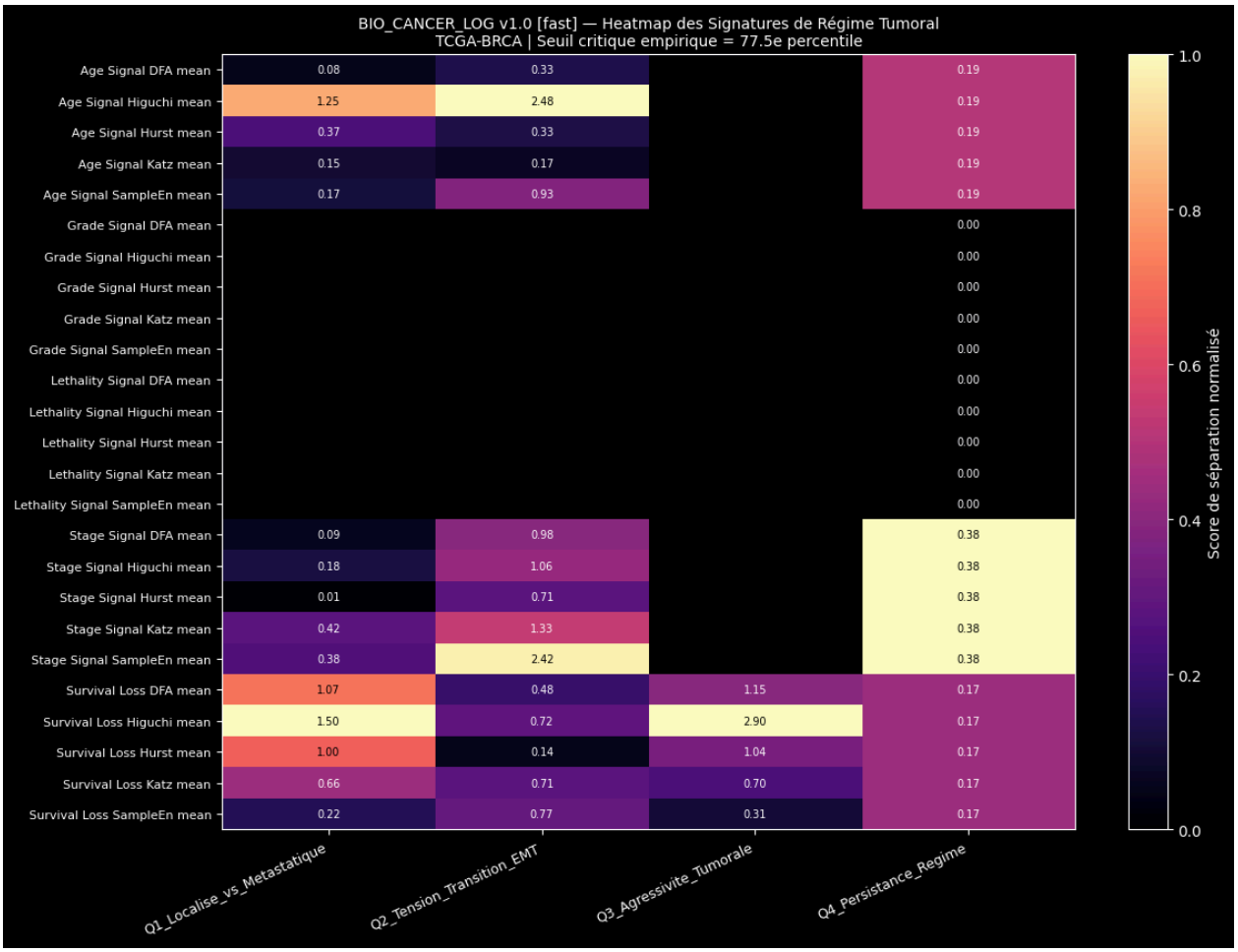
>>> TOP 3 DES SIGNATURES DE RÉGIME PAR QUESTION :

  [Q1_Localise_vs_Metastatique]
    Survival_Loss_Higuchi                        : Score = 1.500
    Age_Signal_Higuchi                           : Score = 1.246
    Survival_Loss_DFA                            : Score = 1.068

  [Q2_Tension_Transition_EMT]
    Age_Signal_Higuchi                           : Score = 2.477
    Stage_Signal_SampleEn                        : Score = 2.418
    Stage_Signal_Katz                            : Score = 1.326

  [Q3_Agressivite_Tumorale]
    Survival_Loss_Higuchi                        : Score = 2.902
    Survival_Loss_DFA                            : Score = 1.146
    Survival_Loss_Hurst                          : Score = 1.039

  [Q4_Persistance_Regime]
    Stage_Signal_DFA                             : Score = 0.379
    Stage_Signal_Hurst                           : Score = 0.379
    Stage_Signal_Katz                            : Score = 0.379
```



[OK] BIO_CANCER_LOG v1.0 terminé.
913 patientes TCGA-BRCA | Seuil EMT empirique : 77.5e percentile